

Siddharth Chatteraj

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EDUCATION

University of California, Santa Barbara: 3.99 GPA

Santa Barbara, CA

B.S. Statistics and Data Science | B.A Theater (Design) | Minor: Language and Speech Technologies *September 2022 - March 2026*
Senior Honors Project: Designing Embodied Wearable Technology to Enhance Kinesthetic Expression for Dance Performance

RESEARCH EXPERIENCE

Computational Psycholinguistics of Listening and Speaking Lab

Santa Barbara, CA

Undergraduate Research Assistant

January 2023 - Present

- Built models of simple dynamical systems of language change in Python by deriving and implementing formulas for statistical model analysis. Optimized parallel process and file writing runtimes from 120 minutes to 30 seconds. Currently writing paper to submit to the Society for Computation in Linguistics and publish in its peer-reviewed proceedings
- Developed an SQL tool to query word frequency and score data from 1000 relational databases and determined best samples by calculating Jensen-Shannon divergence 1000 times for 30 classifications of data using kernel density curves
- Conducted word-based and morph-based phonotactic scoring and analysis of 200000 fake Spanish words using Python Pandas, SRILM, and Morfessor to generate a set of stimuli for an experiment examining Spanish proto-lexicons

CONFERENCE PRESENTATIONS AND EXHIBITIONS

- Ventura, Meredith, Delila Moseley (Lighting and Projection Design by Siddharth Chatteraj). 2025. Curio. *Dance Fest Skopje Opening Night* [performance]. Festival. Macedonian National Theatre, Skopje, Macedonia
 - Developed lighting and projection design grounded in semiotic theory and 1920s–30s cabaret aesthetics to support a dance-theater exploration of embodied trauma. Visual assets were created in After Effects and cued live using QLab
- Chatteraj, Siddharth, Shahil Patel, and Simon Todd. 2024. Deepening understanding of frequency effects in exemplar models of language change [poster]. Conference. *CAMP[7]: 7th Annual California Meeting on Psycholinguistics*. UC San Diego, CA
 - Authored a simulation-based study comparing decay and overwriting architectures in exemplar models using R. Visualized value and variance trajectories across 10,000+ iterations to reveal parameter-sensitive frequency effects
- Chatteraj, Siddharth. 2023. Realms of conception [installation]. *RARE: Realities Altered Realities Emerged*. Exhibition. Santa Barbara Center for Art, Science and Technology, Santa Barbara, CA
 - Designed an interactive audiovisual system using TouchDesigner, Python NLP sentiment analysis, and OSC. Users typed their thoughts to influence color based on emotional tone and adjusted visual scale using physical controls
- Cai, Grant, Henry Ma, Jessie Zhou, Siddharth Chatteraj, Stella Jia. Improving Online Legal Consultation: Strategies for Enhanced Client Assistance [presentation]. *DataFest*. Competition Hackathon. UC Los Angeles, CA. Finalist
 - Evaluated 850000 American Bar Association Free Legal Answers records in 48 hours with Pandas/Natural Language Processing and utilized Python to design charts to propose accessibility improvements
- Nunez, Jesse, Madhav Viswesvaran, Ishaan Avinash, Calais Waring, Siddharth Chatteraj, Anna Lin. 2023. SlangSwitch [presentation and poster]. *UCSB Data Science Project Showcase*. Competition Conference. UC Santa Barbara, CA. 2nd Place
 - Developed a regional slang-to-formal text translator using Styleformer and web-scraped slang corpora. Applied Regex-based tagging and built an interactive demo to demonstrate sociolinguistic style transfer

TEACHING EXPERIENCE

UCSB Department of Computer Science

Santa Barbara, CA

Undergraduate Learning Assistant (ULA)

April 2023 - Present

- Served as Lead ULA, mentoring newer ULAs and coordinating peer teaching strategies across five quarters
- Designed LLM-integrated assessments for one of the first UCSB classes to incorporate generative AI into instruction, introducing students to data science through tools like prompt-based analysis, model-assisted coding, and language model evaluation
- Tutored weekly class sections of 30 beginner data science students in table operations, hypothesis testing, bootstrapping, statistical prediction techniques, linear regression, data visualization, clustering, and classification concepts
- Graded 100-200 weekly coding and analysis assignments and oral presentations and held weekly office hours to provide one-on-one support for conceptual understanding, homework, labs, and projects

Quarter

Course

Role

W24, S24, F24, W25, S25, F25
F23

CMPSC 5A: Introduction to Data Science 1 with Large Language Models
CMPSC 5A: Introduction to Data Science 1

Lead ULA
ULA

S23

CMPSC 5B: Introduction to Data Science 2

ULA

WORK EXPERIENCE

Amazon

Seattle, WA

Business Intelligence Engineer Intern

June 2025 - September 2025

- Saved 92 hours per week of manual analysis by building an internal AI agent that analyzes return cost per unit via SQL and schema analysis and then forecasts global trends with time-series models and ML, informing policy decisions on return centers and backhaul carriers in 3–5 minutes. From learnings, developed two automated Cline AI-agent-creation workflows, enabling engineers to generate working agents in 5 minutes and cutting development time by 20% by starting at testing and refinement
- Investigated unreconciled EU returns by building SQL queries and automating ETL pipelines, analyzing backhaul carrier data to identify fulfillment inefficiencies and automate reporting; Designed QuickSight dashboards tracking weekly, monthly, and quarterly transportation return metrics to support business planning and operational insights

Savie Health*Revenue Manager***Remote/Lompoc, CA***February 2025 - September 2025*

- Tracked and forecasted revenue from Medi-Cal and grants to support financial reporting and sustainability planning

*Data Analyst**August 2024 - February 2025*

- Built time series models in Python to forecast patient attendance based on potential operating changes and analyzed data to help the organization earn over \$65,000 in grants; Organized and managed donor and patient databases
- Rebranded organization's website using WordPress and assisted leadership with outreach communication

*Data Analysis/Marketing Intern**July 2022 - August 2024*

- Created promotional graphics and e-blasts in English and Spanish to advertise the startup nonprofit and helped secure over \$100,000 in funding through email marketing and data reports for grant requests
- Built and managed Instagram, Facebook, and LinkedIn pages, which receive an average of 1500 monthly engagements

San Francisco District Attorney's Office**San Francisco, CA***San Francisco Sentencing Community Initiatives Data Fellow**June 2023 - September 2023*

- Created a regeneratable temporal and geospatial data memo for distribution to community members across the city using HTML, JavaScript, and Python Plotly to visualize crime trends in San Francisco supervisor and police districts
- Analyzed several databases of over 200000 county crime records and visualized crime location trends using Geopandas, R, Python Plotly, and ArcGIS for grant requests. Wrote a 21-page guide on how to analyze and map geospatial data

Daily Nexus**Santa Barbara, CA***Data Editor and Games Section Founder**April 2023 - Present*

- Oversaw the production of over 65 data-driven articles and mentored more than 200 writers on data storytelling, sourcing, and visualization, leading to multiple statewide awards: 1st Place for Best Multimedia Package (2025), 2nd Place for Best Interactive Graphic (2024), and 1st Place for Best Infographic (2023) from the California College Media Association
- Founded the Games section and launched its website, making the Nexus one of the first college newspapers with a dedicated games page. Designed interactive crosswords with JavaScript and taught workshops on data visualization, NLP, and web scraping
- Led management of The Labyrinth, a public dashboard visualizing 16 years of UCSB course data—over 40,000 classes—using D3.js. The tool is now accessible to UCSB's 30,000 students to explore grade distributions by instructor, quarter, and department

*Assistant Data Editor**September 2022 - April 2023*

- Self-taught Python to create multiple interactive charts and interviewed sources to compose articles about events across Santa Barbara. Wrote and edited articles on the UC strike, Diwali, Halloween, historical crime trends, sports, and D.E.I.

KCSB FM 91.9**Santa Barbara, CA***News Anchor and Reporter**September 2022 - September 2023*

- Anchored weekly newscasts and investigated major local stories including the Cuyama water conflict, Isla Vista housing crisis, and cliff fatalities by interviewing local residents and public officials. Edited segments in Audacity for broadcast

THEATER DESIGN EXPERIENCE

Role	Production	Company	Date
DESIGN:			
Assistant Projection Designer	<i>TrollFest¹</i>	Ray Troll and the Ratfish Wranglers	05/2025
Technical Director, Lighting Designer, Projection Designer, Stage Manager	<i>UCSB Dance Company European Tour²</i>	UCSB Dance Company	04/2025
Lighting Designer	<i>Afterlight</i>	UCSB Division of Dance	04/2025
Projection Designer, Stage Manager	<i>Refractions: Voices in Light & Dark</i>	UCSB Dance Company	03/2025
Assistant Lighting Designer, Lead Technician	<i>Strange Birds</i>	LAUNCH PAD	02/2025
Lighting Designer, Stage Manager	<i>UCSB Dance Company Fundraiser</i>	UCSB Dance Company	02/2025
Projection Designer	<i>The Threepenny Opera</i>	UCSB Division of Theater	11/2024
Projection Designer, Lead Technician	<i>Kinetic Lab</i>	UCSB Division of Dance	06/2024
Assistant Projection Designer	<i>Indecent</i>	UCSB Division of Theater	05/2024
STAGE MANAGEMENT:			
Stage Manager	<i>Intersections: Faculty Dance Concert</i>	UCSB Division of Dance	11/2024
Stage Manager	<i>Transcend</i>	UCSB Division of Dance	04/2024
Stage Manager	<i>The Crucible</i>	UCSB Division of Theater	03/2024
Stage Manager	<i>Waiting</i>	UCSB Division of Theater	05/2023
TECHNICIAN:			
Lead Technician	<i>Senior Dance Solos</i>	UCSB Division of Dance	06/2025
Lead Technician	<i>Los Lazos Que Unen</i>	Santa Barbara Dance Theater	10/2023
Lead Technician	<i>The Last Days of Judas Iscariot</i>	UCSB Division of Theater	06/2023
Lead Technician	<i>Sound and Smoke</i>	UCSB Division of Dance	04/2023
Lead Technician	<i>Night Night, Roger Roger</i>	UCSB Division of Theater	03/2023

¹Concert produced in Juneau, Alaska²Tour produced in Santa Barbara, CA; Barcelona, Spain; Nowy Sącz, Poland; Tarnow, Poland; Sofia, Bulgaria; Skopje, Macedonia